

EXPERIMENT 20: TOWERS OF HANOI

Aim

To implement the Towers of Hanoi problem using Python.

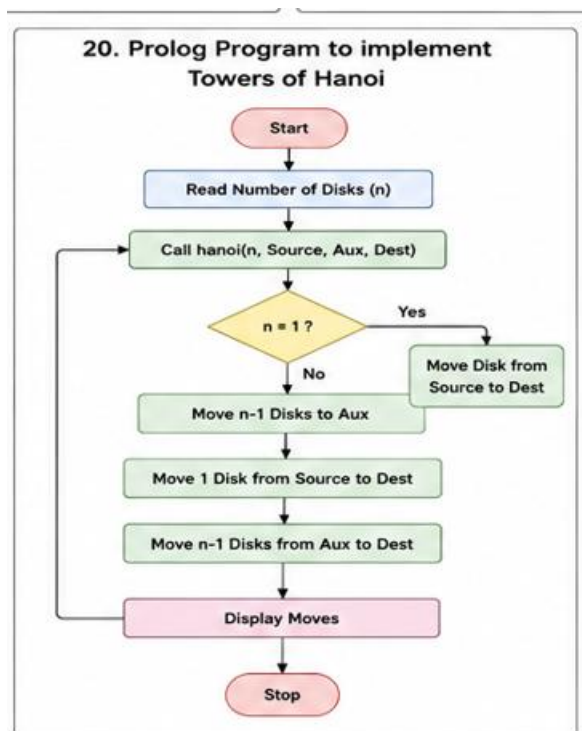
Objective

To move all disks from the source rod to the destination rod using recursion.

Algorithm

1. Start.
2. Read the number of disks.
3. If there is only one disk, move it directly.
4. Move N-1 disks from source to auxiliary.
5. Move the largest disk from source to destination.
6. Move N-1 disks from auxiliary to destination.
7. Display all moves.
8. Stop.

Flowchart



Code

```
def hanoi(n, source, auxiliary, destination):  
    if n == 1:  
        print("Move disk from", source, "to", destination)  
        return  
  
    hanoi(n - 1, source, destination, auxiliary)  
  
    print("Move disk from", source, "to", destination)  
  
    hanoi(n - 1, auxiliary, source, destination)  
  
n = int(input("Enter number of disks: "))  
  
hanoi(n, "A", "B", "C")
```

Output :

```
Enter number of disks: 3  
Move disk from A to C  
Move disk from A to B  
Move disk from C to B  
Move disk from A to C  
Move disk from B to A  
Move disk from B to C  
Move disk from A to C
```

Result

The Towers of Hanoi problem was successfully implemented using Python.

Conclusion

The program demonstrates recursion by solving the Towers of Hanoi problem and displaying the sequence of disk movements.